



Marie Roth

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Born 18 April 1999

EDUCATION

Apr. 2022 – now

PhD in Mathematics

RPTU Kaiserslautern–Landau (Germany) and Aix-Marseille University (France)

- Thesis title: *Characters sheaves and modular representations of finite reductive groups*
- Supervisors: Olivier Dudas and Gunter Malle
- Manuscript hand in in Dec.2024, defense planned on the

2020 – Feb. 2022

Master's degree in Mathematics

Ecole Polytechnique Fédérale de Lausanne (Switzerland)

- Thesis title: *Ennola duality in subgroups of classical groups*
- Supervisors: David Craven and Donna Testerman
- Final grade: 5.83/6 (6/6 Master's thesis)

2017 – 2020

Bachelor's degree in Mathematics

Ecole Polytechnique Fédérale de Lausanne (Switzerland)

- Third year in exchange at the University of Waterloo, Canada.
- Thesis title: *On the finite relatedness of term clones of finite groups*
- Supervisor: Ross Willard
- Final grade: 5.31/6 (6/6 Bachelor's thesis)

2014 – 2017

Baccalauréat Scientifique

Lycée Présentation de Marie (France)

- Final grade: 19.75/20 "mention très bien" and magna cum laude

2011 – 2017

Euler class graduate

Ecole Polytechnique Fédérale de Lausanne (Switzerland)

- Intended for children with a very high mathematical intuition. Cover the whole maths program until the end of the first year at EPFL.

EMPLOYMENT

Apr. 2022 – Apr.2025

Research Assistant

RPTU Kaiserslautern–Landau (Germany)
with Gunter Malle

- funded by SFB–TRR 195 "Symbolic Tools in Mathematics and their Application" of the German Research Foundation (DFG)

July 2021

Auxiliary Staff

Hospital Collonges-Bellerive (Switzerland)

2020 – 2021

Teacher

Lycée Présentation de Marie (France)

- 2018 – 2020 **Teaching assistant**
Ecole Polytechnique Fédérale de Lausanne (Switzerland)
• Linear Algebra, Physics, Consolidation (Mise à Niveau), Information Computation and Communication, Lie Algebras, Mathematical Logic.
- 2014 – 2019 **Tutor**
Ecole Polytechnique Fédérale de Lausanne (Switzerland)
• Tutored for Cours Euler (exercise classes).
- Jul. 2017 **Euler Camp Counsellor**
Ecole Polytechnique Fédérale de Lausanne (EPFL)

TEACHING AND ORGANISATIONAL EXPERIENCE

- 2022–now Member of the gender committee of the SFB–TRR 195
- 2023 Exercise class for Representation Theory held by Gunter Malle
- 2023 Co-organised the workshop “How to advance your research career and get funding” in Kaiserslautern (Germany)
- 2023 Exercise class for Character Theory of Finite Groups held by Caroline Lassueur
- 2022 Exercise class for Representation Theory held by Caroline Lassueur
- 2021 Organised a reading group with other students to learn about the representation theory of semisimple Lie algebras, with the help of Donna Testerman.

PUBLICATIONS AND PREPRINTS

M. Roth. *Unitriangularity of decomposition matrices of the unipotent ℓ -blocks for simple adjoint exceptional groups*. 2024. arXiv: 2402.17616 [math]

TALKS

Invited talks

- Nov. 2024 On the unitriangularity of decomposition matrices of finite reductive groups, “Groups and Representation Theory in Lausanne 2024” (Switzerland)

Contributed talks and posters

- Dec. 2024 Poster: Are decomposition matrices of finite groups of Lie type unitriangular? “Women in Algebra and Symbolic Computation III” in Bad Dürkheim (Germany)
- Sep. 2024 Poster: Are decomposition matrices of finite groups of Lie type unitriangular? “Groups and representations: after Roger Carter” in Warwick, (United Kingdom)
- Jul. 2024 On the unitriangularity of decomposition matrices of finite reductive groups, “Group Theory in Florence IV” (Italy)
- Mar. 2024 Why are exceptions so valuable: a quick introduction to the Moonshine conjecture, “Séminaire doctorant CPT/I2M à Luminy” (France)
- Oct. 2023 Character sheaves of the principal series restricted to a conjugacy class, “Representation Theory at Villa Denis” near Kaiserslautern (Germany)
- Sep. 2023 Poster: Are decomposition matrices of finite groups of Lie type unitriangular? Annual Meeting of the SFB–TRR 195 in Saarbrücken (Germany)
- Dec. 2022 Unitriangularity of Decomposition Matrices for Exceptional Simple Adjoint Groups of Lie Type, “Nikolaus Conference 2022” in Aachen (Germany)

Working groups seminars

- 2024 Unipotent characters, Jordan decomposition of characters, *Oberseminar: Deligne–Lusztig Theory*, RPTU Kaiserslautern (Germany)
- 2024 Théorème de Kneser, *Groupe de travail: Cohomologie Galoisienne*, AMU (France)
- 2023 Thompson factorisation, *Oberseminar: Selected Topics in Group Theory*, RPTU Kaiserslautern (Germany)

- 2023 Normalisers and centralisers, *Oberseminar: Fusion Systems*, RPTU Kaiserslautern (Germany)
- 2022 Quivers, representations, path algebra I, *Oberseminar: Quiver Representations and Quiver Varieties*, RPTU Kaiserslautern (Germany)
- 2022 Unipotent classes and Kawanaka representations, *Oberseminar: Representation Theory of Finite Reductive Groups*, RPTU Kaiserslautern (Germany)

CONFERENCES AND SUMMER SCHOOLS

- Dec. 2024 "Nikolaus Conference 2024" in Aachen (Germany)
- Sep. 2024 "Representation Theory at Villa Denis" near Kaiserslautern (Germany)
- Jul. 2024 "Joint Block Seminar 2024 on Algebraic Number Theory" in Aachen (Germany)
- Mar. 2024 "Colloque tournant du Réseau Thématique «Algèbre»" in Marseille (France)
- Sep. 2023 IRTG Block seminar on Category Theory in Trippstadt (Germany)
- Aug. 2023 "LMS-Bath Summer School and Symposium on Geometric and Categorical Representation theory" in Bath (United Kingdom)
- Jul. 2023 "LMS Research School: Algebraic Groups and Their Representations" in Birmingham (United Kingdom)
- Jun. 2023 "Representations of Reductive Groups" in Aachen (Germany)
- Mar. 2023 IRTG (Integrated Research Training Group of the SFB-TRR 195) Block Seminar at the Ebernburg, Bad Münster (Germany)
- Oct. 2022 "Darstellungstheorietage 2022" in Kaiserslautern (Germany)
- Sep. 2022 "Representation Theory at Villa Denis" near Kaiserslautern (Germany)
- Sep. 2022 "Young Group Theorists Workshop" in Les Diablerets (Switzerland)
- May 2022 Workshop "Counting conjectures and beyond" in Cambridge (United Kingdom)

ADDITIONAL SKILLS

Languages

French (Mother tongue), English (Fluent PCE C2), German (Academic level B1)

Programming

$\text{\LaTeX}$ (advanced), GAP (intermediate), Python (beginner), C++ (beginner), Matlab (beginner)

References

Gunter Malle, Fachbereich Mathematik, RPTU Kaiserslautern, Postfach 3049 67653 Kaiserslautern, malle@mathematik.uni-kl.de
 Olivier Dudas, Institut de Mathématiques de Marseille - I2M Campus de Luminy, Avenue de Luminy, Case 930, 13288 Marseille Cedex 9, olivier.dudas@univ-amu.fr